Section 3: Technical Evidence

2024 analysis: 79 complete-case Ortsteile from 88 workbook areas; no missing value imputation.

Three indicators: unemployment quotient, social-insurance employment participation, and SGB II benefit receipt under age 65. Percentages have different denominators.

Highest composite scores: Ohlenhof, Gröpelingen, Grohn, Blumenthal, Lindenhof. Highest TOPSIS closeness: Ohlenhof, Gröpelingen, Blumenthal, Lindenhof, Tenever.

Rank Spearman correlation: 0.9743. K-means selects 2 nominal profiles; silhouette 0.5277.

Scores are relative to the covered sample and describe partial disadvantage. They do not establish individual poverty, causal effects, current 2026 conditions or an allocation rule.

Historical change claims remain blocked. Boundary/definition history and exact current reference dates need municipal verification. Income context is kept at 2020.

Architecture review: current local files and Python pipeline are implemented. HTTPS acquisition, database/cloud hosting and a planner dashboard are proposed. Sensors and their networks are only future extensions.

Technical evidence for group review; not a final university report section.
Source: supplied municipal workbook. Primary reporting year: 2024.

Bremen Smart Governance: Four-Layer Blueprint

Local implementation and proposed deployment are explicitly distinguished

1. Sensing / Data Acquisition

CURRENT: municipal XLS + historical CSV; JSON catalogue
PROPOSED: versioned municipal release retrieval; no live sensors

2. Transmission / Network

CURRENT: local filesystem transfer
PROPOSED: HTTPS / supported secure API; IoT networks only as future extensions

3. Processing / Cloud or Edge

CURRENT: Python validation, CSV storage, notebooks; index / TOPSIS / K-means
PROPOSED: scheduled jobs, PostgreSQL, immutable object store / municipal hosting

4. Application / AI and Decision Support

CURRENT: static tables, charts, PDFs and evidence packages
PROPOSED: planner dashboard; weights, nominal profiles, provenance, exclusions

Validation gates: primary annual coverage passed; historical change gate blocked.
Proposed decision support requires human review; no operational dashboard deployed.